

- `>>> lst = [1, 2, 3, 4, 5]`
- `>>> lst.append(6)`
- `>>> print(lst)`
- `[1, 2, 3, 4, 5, 6]`
- `>>> lst.extend([7,8])` #extend the list with another iterable
- `>>> print(lst)`
- `[1, 2, 3, 4, 5, 6, 7, 8]`

Insertions in the middle of the list can be performed using the function, `insert(index,element)`

- `>>> lst.insert(2, 9)`
- `>>> print(lst)`
- `[1, 2, 9, 3, 4, 5, 6, 7, 8]` #insert element 9 at index (2)

- Removing elements from a list can be done in multiple ways, if we wish to catch the element that is getting deleted we can use the `pop()` function of list module which takes the index as a parameter and removes the element at that index. If the index is not supplied, it pops the last element from the list.

In case we want to perform a deletion without catching the element being deleted, we can use the `del` operator. It is important to take note of the fact that `pop` is a function while `del` is an operator.

- `>>> a = [1, 2, 3, 4]`
- `>>> a.pop(2)`
- `3` #returns the element being deleted
- `>>> print(a)`
- `[1, 2, 4]`
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- `>>> a = [1, 2, 3, 4]`
- `>>> del a[2]`
- `>>> print(a)`
- `[1, 2, 4]`

Both `pop` and `del` are index-based operations i.e they delete an element by its index and not by its value.